

Three potential directions for 3rd-year paper

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Goals

1. Goal for this document: summarize work on three ideas, and determine a topic for my third-year paper.

Summary

I have organized these ideas in descending order of how promising I think they are. As an intermediate goal, I would like to develop one of these to a point where I can present at SRES in the autumn quarter (i.e. by November 2025).

- The gambling idea was my proposal for Evan Rose's class, and I have done some basic analyses with the SafeGraph data for it. Evan seems pretty excited about me continuing it.
- The senior migration idea came out of Josh Gottlieb's class and is the least developed, but I think it is also in a less-crowded area than the other two ideas.
- I discussed the partisanship and zoning idea in the 3rd-year seminar in January. The fact that I have spent a lot of time on this without finding a more convincing empirical strategy is why I am most pessimistic on it.

1 Are land-based and online gambling substitutes or complements?

1. **Motivation: why is this an interesting question? What do you have in mind here?** Since the Supreme Court's *Murphy v. NCAA* decision on May 14, 2018, which struck down the 1992 Professional and Amateur Sports Protection Act (PASPA), **38 states and D.C. have legalized sports betting.**¹ Americans wagered **\$119.84 billion on legal sports books in 2023** and **\$147.91 billion in 2024**. While the policy legalized all forms of sports betting, its primary effect in practice was the legalization of online sports betting; **more than 95%** of all legal sports bet dollars in 2024 were wagered online. On the other hand, gambling has negative externalities. However, this does not mean legalization is inefficient; it could be reasonable for a social planner to allow it, and tax it. But is this newly legalized activity having its desired fiscal effects? If not, states might be legalizing a harmful activity without getting much net tax revenue in return.

¹Nevada received an explicit carveout in the 1992 PASPA, so was not affected by the Supreme Court ruling

2. **Advance: what is the key novelty here, how will you advance compared to the existing literature?**

Empirical: This paper would be the first to answer the research question with national data; (Humphreys, 2021) asks a similar question in West Virginia but with a limited sample and time period. The author also does not attempt to estimate any preference parameters; it is a purely empirical paper.

Theory: No obvious novelty here yet, compared to, e.g., (Miller and Seo, 2021) or (Gentzkow, 2007); though I will probably modify one of model to better match the setting.

3. **Literature: Some key papers papers, up to five, one suffices. Discuss them briefly.**

This idea is closest to (Humphreys, 2021), which studies the effect of legalizing sports gambling in West Virginia. The period studied is September 2019 - March 2020, and the sample includes only 5 casinos. The author uses weekly NFL and NBA games as an instrument that shifts demand for sports betting, and argues legalization led to large, negative fiscal effects. In the context of cannabis legalization, (Miller and Seo, 2021) estimate legalization's effect on the demand for alcohol and tobacco products. They find positive net fiscal effects from legalization even though purchases of alcohol and tobacco both fell. (Gentzkow, 2007) asks a similar question for newspapers, and develops a new demand estimation model.

4. **An outline of your approach. What do you plan to do. It should be, in essence, a research plan that you can next sit down and carry out.**

- Empirical: Brainstorm other analyses I could do with the SafeGraph data that have either more power, less pre-trends or both (e.g. analyzing one state's legalization at a time). Raw means of in-person casino traffic, created with the SafeGraph data are documented in 1.
- Empirical/Data: See if there is existing micro-data on gambling behavior (or a way to proxy for this with the TransUnion data).
- Empirical: Start designing a survey to collect this data. (Gentzkow, 2007) needs survey data for his study of newspapers; Evan Rose also suggested this as a next step, so I predict the final product will require survey data.
- Institutional details: Understand which activities at a land-based casino might be complements versus substitutes for online sports gambling
- Institutional details: Code the state's tax revenue as a function of various forms of gambling revenue for one major state. Even if casino revenue declines, a lot of the tax revenues are fixed (e.g. casinos in Florida pay \$1M+ annual license fee regardless of how much they earn)
- Literature: Read (Gentzkow, 2007) closely
- Theory: Review notes on demand estimation to inform a structural model. Start writing structural model.

5. **A guess as to what you will find.** I do not expect this question will have a one word answer. The main nuance I expect to find is that legalization induces more land-based gambling for people that live near casinos, but lowers land-based gambling from people who live further away.

1.1 My thoughts

1. Hours spent on this so far: ≈ 35
2. There are still two questions being asked; depending on data, etc., I might focus more on one or other, a policy question : **Has legalizing sports gambling led to positive fiscal effects, or has it merely cannibalized in-person gambling?** , and a more economic question: **Are online sports betting and in-person gambling substitutes or complements?**
3. Perhaps the most obvious / highest risk of getting scooped? If I do it, I should move quickly.

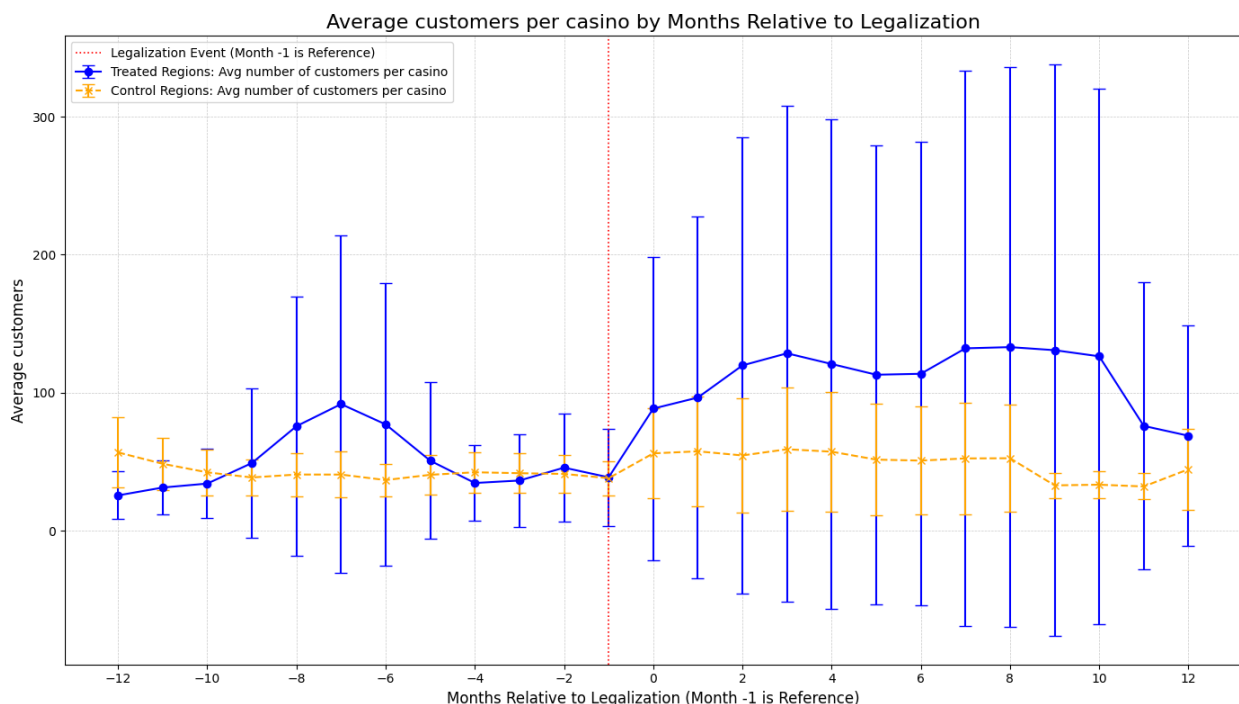


Figure 1: SafeGraph data on number of customers visiting casinos which legalized or not. An observation is a casino-month. Restricts to establishments with customer counts in 80%+ of months and present in both Feb 2019 and July 2024. Standard errors clustered at the state level. Bars indicate 95% confidence intervals. Treated sample consists of 9 unique casinos in 5 states (546 casino-months). Control sample consists of 17 unique casinos in 9 control states (1,033 casino-months). A casino is treated if sports gambling was legalized in that state. Treated states are AZ, MI (2021), LA, NY (2022), OH (2023). Controls are AL, AK, CA, ID, MN, MO, OK, TX, and WI.

2 Is senior migration driving up rents?

1. **Motivation: why is this an interesting question? What do you have in mind here?**
Approximately 3 million seniors move each year, $\approx 1/3$ of which are interstate moves, and bring with them federally funded healthcare and social security dollars. We do not know the effects of these population flows on rents or tax revenue, or whether municipalities should be trying to attract them or induce them to leave. While senior migration is a small share of overall migration, the 55+ population hold a majority of the US net worth, and where they live determines where a large share of government spending is funneled, with social security and Medicare dollars combined totalling $\approx 35\%$ of federal spending.
2. **Advance: what is the key novelty here, how will you advance compared to the existing literature?** (1) This would be the first paper to study the effects of senior migration on rents, and tax revenues, in the US. However, (Badilla Maroto et al., 2024) ask and answer similar questions in the French context with very good micro data. (2) This paper will help uncover why retirees move; for example, preliminary regressions (see Table 1) show that local climate and population density are each independently correlated with changes in the elderly population share, even after considering various income and housing affordability measures. Mechanisms like this are not explored in the literature.² (3) The fiscal externalities of retiree migration are unclear; and it's not clear what we learn about the US from the French setting, where property taxes are the primary source of local revenue vs. in the US where sales and income taxes are more important.
3. **Literature: Some key papers, up to five, one suffices. Discuss them briefly.** This idea came out of a disagreement with some conclusions from (Dingel, Gottlieb, Lozinski, and Mourot, 2023); in this model, more medical services in a place are an unalloyed good (due to agglomeration effects in medical quality); but of course some of these medical amenities are capitalized into rents. The closest paper to this idea is a working paper by (Badilla Maroto, Faber, Levy, and Muñoz, 2024) study the question in France with very good micro data and find a positive causal effect on house prices and various sources of tax revenue, using a shift-share design. Another paper with Muñoz as an author studies international tax migration of seniors, (Kalin, Levy, and Muñoz, 2024). A 2022 job market paper, (Komissarova, 2022) introduces a model of endogenous amenity formation in response to age composition, with a focus on explaining changes in retirees' migration patterns since 1960.
4. An outline of your approach. What do you plan to do. It should be, in essence, a research plan that you can next sit down and carry out.
 - Empirics: Using public data, try to get a better understanding of what's correlated with senior migration; i.e. try to get a higher R^2 in a regression with senior migration on the left-hand side (i.e. conceptually extend table 1).
 - Institutional details: Find how common it is for cities/areas to be trying to attract seniors.
 - Literature: Go through (Badilla Maroto et al., 2024) and see which analyses I could do with public data in the US.

²Local climate is proxied by The USDA's 1999 natural amenities scale, which combines six measures of climate, topography, and water area that reflect environmental qualities most people prefer (warm winters, winter sun, temperate summers, low humidity, topographic variation, and water area). A simple additive scale was developed, with some adjustment for the interrelationships among the measures, standardizing scores for each county in the lower 48 states.

- Literature: Try to understand why ([Komissarova, 2022](#)) did not publish so well / or do well on the market.
 - Data: Apply for Medicaid / Medicare data which we have access to here at Chicago.
 - Theory: Write a simple static model to understand and clarify key forces.
5. **A guess as to what you will find:** I anticipate seniors will have positive fiscal externalities and positively impact local housing prices. In a welfare calculation: this will positively impact local home-owners, but negatively impact renters.

My thoughts / other feedback I have gotten

1. Hours so far: ≈ 25
2. I think the biggest threat here is lack of good US data for certain measures, and the similarity to ([Badilla Maroto et al., 2024](#)).
3. Spoke with Josh Gottlieb about this twice

Table 1: Correlates of County-Change in Elderly Population Share (2000-2010)

	(1)
Constant	-1.0159 (1.2866)
Share 65+ (2000 Levels)	-0.0206 (0.0154)
Log Population (2000)	-0.3729*** (0.0505)
Median HH Income (2000), thousands	0.1725*** (0.0256)
Median House Value (2000), tens of thousands	-0.2684*** (0.0610)
Median Rent (2000)	-0.0076*** (0.0027)
Top State Income Tax Rate (2000)	-0.0054 (0.0240)
Healthcare Employment Share (2000)	-2.9147*** (0.4840)
USDA Natural Amenities Scale (1999)	0.0884** (0.0385)
Median House Value to Median Income Ratio (2000)	18.8656*** (3.0379)
Median Rent to Median Income Ratio (2000)	0.0000 (0.0000)
Land Area (sq. miles) (2000)	0.0000 (0.0000)
Population Density (2000)	-0.0002*** (0.0000)
Observations	3,098
R^2	0.257
Adjusted R^2	0.254
States (clusters)	48

3 What is the causal effect of Democrats on housing regulations and affordability?

1. **Motivation: why is this an interesting question? What do you have in mind here?**
This idea is motivated by facts: (1) There is a cross-sectional correlation between housing regulations and share of Democrats in an area, as most recently documented by (Bartik, Gupta, and Milo, 2024) at the municipality-level. This has led several influential pundits and economists to argue that liberal cities are especially ineffective at building housing, often implying liberal ideology itself has a negative causal effect on housing supply. However, it is not clear this relation is causal - e.g. see the long differences analysis regression in Fig. 2 showing that changes in regulation and changes in the Democrat share are actually negatively correlated. (2) People have been leaving blue states and migrating to red states, as shown in Fig. 3. How much of this trend can be explained through Democrats making housing more expensive?
2. **Advance: what is the key novelty here, how will you advance compared to the existing literature?** Prior work in economics and political science relies on close mayoral races, or focuses only on California. There is no paper studying the effect of liberal ideology on housing supply, zoning regulations, or housing supply elasticities at the national level. See below for further discussion.
3. **Literature: Some key papers papers, up to five, one suffices. Discuss them briefly.** (de Benedictis-Kessner, Jones, and Warshaw, 2024) and (Ferreira and Gyourko, 2023) both use regression discontinuity design (RDD) evidence, with close mayoral elections giving random variation for which party holds power. The outcome variable is housing units built. (de Benedictis-Kessner et al., 2024) finds that electing a Democrat mayor has a small positive significant effect on multi-family supply, but not total units. However, an RDD primarily uses variation near 50%. This is a major drawback of this paper, since very liberal cities are where affordability is the greatest problem (e.g., New York City, Boston, San Francisco, Los Angeles). There is also a disconnect between the evidence emphasizing mayors' party and the institutional context. Zoning decisions are primarily determined by local community groups (e.g. Save Berkeley's Neighborhoods), zoning commissions, and city council, rather than mayors. Thus, any paper studying the effect of ideology on housing supply should focus on residents' ideology, not only the mayor's. (Kahn, 2011) is another related paper. His main specifications regress housing permitted in California cities on liberal vote share (Green Party + Democrats). This is a continuous measure, so, unlike the RDD evidence, this paper can answer questions about the degree of ideology. The author only looks at California and attempts to deal with endogeneity by using lagged liberal vote share as an IV for liberal vote share, and by using long differences.
4. **An outline of your approach. What do you plan to do. It should be, in essence, a research plan that you can next sit down and carry out.**
 - The ideal finished product would (i) include causal effects of Democrats on zoning (or some related measure), and (ii) a quantitative model with rents, migration, partisanship, and endogenous housing supply restrictions, which would let me measure the importance of the Democrat $\rightarrow$ zoning mechanism for explaining interstate migration flows.

- Empirics: Keep looking for an instrument / empirical strategy to identify “causal effect of democrats on x ” where x may be zoning.
 - Empirics: Look more deeply at whether I can use close election evidence. Milena Almagro did mention she couldn’t find any close election in Chicago.
 - Empirics: Try to find the longest panel of zoning regulations I can. A lot of Milena’s ideas feedback in 3rd-year seminar in January required having panel data to implement (e.g. structural break analyses).
 - Empirics: Try doing an analysis with building permit survey on the left-hand side; this would give me a panel but its interpretation is less straight-forward.
 - Model: I currently have a static spatial model with two types, one which prefers zoning more than the other. This is ad hoc, so I now need to think about what might let me endogenize the relationship between partisan preferences and housing regulation.
 - Literature: Look at models of cultural assimilation to think about how party might endogenously change over time.
5. **A guess as to what you will find.** I anticipate that the effect of Democrats’ share on zoning is probably smaller than suggested by naive cross-sectional regressions, but that there is a positive causal effect. If there is an effect, I would anticipate that it operates through support of unions, environmentalism, and a preference for mandating affordable housing rather than deregulation. I still do not have a strong prior on what I expect to find in the model, or whether the paper needs a model.

My thoughts / other feedback I have gotten

1. Hours so far: ≈ 70
2. This is the project I presented in January at 3rd-year seminar. Good variation seems to be the binding constraint here, and the thing I have made the least progress on; e.g. need exogenous variation in Democrats in an area.
3. I’ve spoken with Milena about this a few times and her advice is to see if I can find a causal effect of Democrats on zoning while working on the model in parallel.

References

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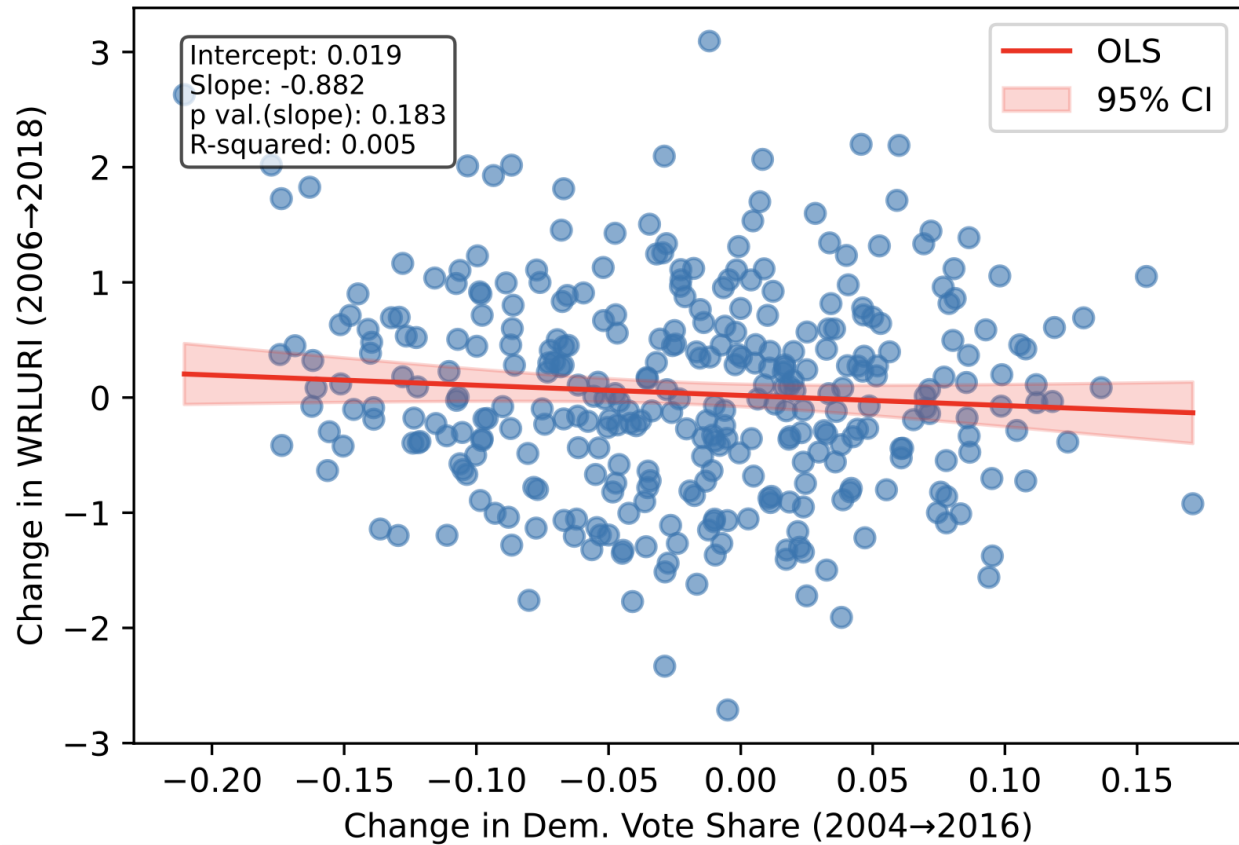


Figure 2: WRLURI Change (2006→2018) vs. Dem. Vote Share Change (2004→ 2016). The WRLURI has only been conducted twice and the change is from 2006 to 2018. I create a weighted average of the WRLURI at the county-level, and observations are the 342 counties in both the election data and both WRLURI waves.

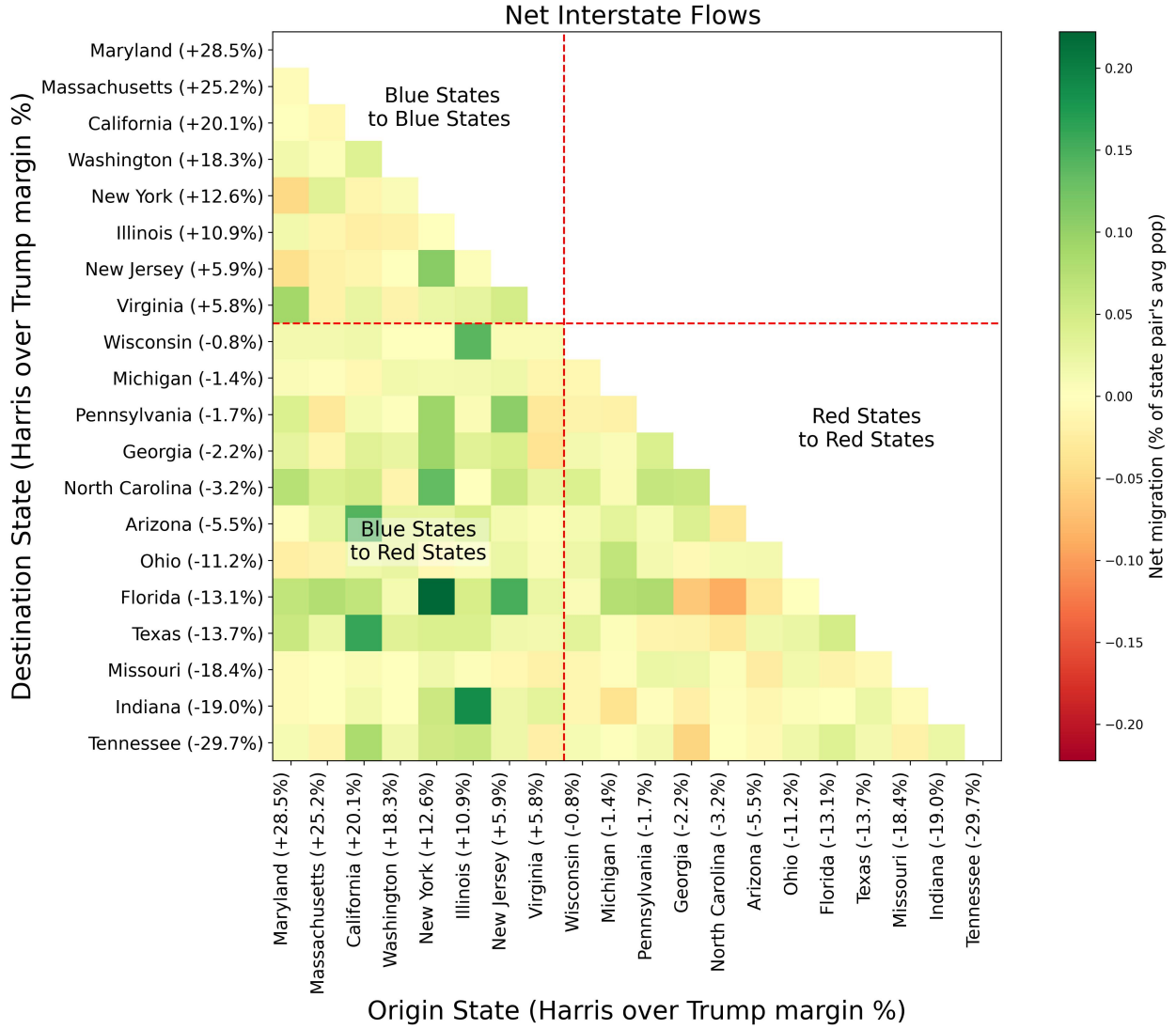


Figure 3: Cells colored based on the following measure of relative net flows: $\frac{gross\ flows_{o \rightarrow d} - gross\ flows_{d \rightarrow o}}{0.5(Pop_d + Pop_o)}$, so a green cell means that, on net, many people moved from (that column) to (that row). An orange/red cell means many people moved from (that row) to (that column). Sample is 20 most populous states, representing about 70% of the US population.

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